Analysis, Design, and Fabrication of a Multimodal Instrumented Prosthetic-Liner Slab

1 Introduction and Background

2 Literature Review

2.1 The prosthetic socket–residual-limb interface as a sensing problem

The mechanical and thermal conditions at the interface between a residual limb, liner, and prosthetic socket strongly influence comfort and tissue health. The liner must simultaneously suspend the prosthesis, distribute load, limit relative motion, and protect soft tissue. These functions create a difficult measurement environment. Normal pressure is produced as load is transferred between the limb and socket, while tangential motion produces shear. The liner also forms a comparatively closed thermal and moisture barrier. Heat and perspiration can therefore accumulate during activity, changing both the condition of the skin and the mechanical behaviour of the interface. A useful instrumented liner should consequently measure more than one isolated variable. Normal pressure, shear, temperature, and humidity represent complementary aspects of the same physical problem.

The importance of this combined view is supported by the broader interface- measurement literature. Al-Fakih *et al.* reviewed techniques used to measure socket-interface stresses and showed that pressure has historically received more attention than shear, even though both contribute to the mechanical state of the residuum [1]. Commercial pressure films and discrete transducers can provide useful measurements, but their thickness, wiring, calibration, and sensitivity to curvature can disturb the interface being measured. Shear measurement is more demanding because a sensor must separate tangential loading from normal compression while remaining sufficiently thin and compliant. This is a central design constraint for the present work: the sensing system must not create a local prominence that changes the load distribution or becomes a new source of discomfort.

Thermal and moisture measurements introduce a related set of constraints. Klute *et al.* demonstrated that prosthetic socket and liner materials form barriers to heat transfer [2]. The consequences are not only subjective warmth. Increased temperature promotes perspiration, while trapped moisture may alter friction, reduce suspension, soften the skin, and contribute to irritation. Paternò *et al.* accordingly argue that temperature and humidity should be studied alongside mechanical loading rather than treated as secondary comfort variables [7]. The literature therefore supports a multimodal architecture; however, it also shows that the utility of such an architecture depends on integration. A sensor that is accurate in isolation is not necessarily suitable when embedded in a flexible liner.

2.2 Multiaxial capacitive sensing: Albrecht *et al.*

Albrecht *et al.* developed an inkjet-printed capacitive sensor intended to distinguish normal and two-directional shear loading [3]. The design is especially relevant because it addresses the cross-sensitivity problem without relying on a thick, rigid multi-axis load cell. Four capacitors, denoted C_1–C_4, are formed by patterned conductive layers separated by a compliant dielectric. The electrodes use an E-shaped geometry. Multiple electrode "fingers" increase the edge length over which overlap changes when one conductive layer moves relative to the other. The initial overlap is arranged near the middle of its working range, allowing the capacitance of opposing elements to increase or decrease with the direction of shear.

The four-capacitor arrangement provides a physical basis for resolving the loading components. Under compression, the dielectric thickness changes and the capacitors exhibit a broadly common-mode response. Under shear in one direction, one opposed capacitor pair responds differentially; the orthogonal pair performs the same function for the second tangential direction. Consequently, sums and differences of the four outputs can be used to estimate normal force and signed shear in two axes. This is a more appropriate model of the prosthetic interface than a pressure-only sensor because socket loading is not purely normal. Pistoning, rotation, and gait-related relative motion can produce tangential forces even where normal pressure appears acceptable.

The reported device used inkjet-deposited conductive traces approximately 412 nm thick and a compliant dielectric approximately 40 µm thick. The electrode fingers were approximately 210 µm wide with a 40 µm gap, and each finger was approximately 8 mm long. Over an investigated range of approximately 0.1 N to 8 N, Albrecht *et al.* reported a normal-force sensitivity of about $5.2\,\mathrm{fF\,N^{-1}}$ and a shear-force sensitivity of about $13.1\,\mathrm{fF\,N^{-1}}$ [3]. The differential shear response was direction-dependent, while the four-element arrangement could increase the combined output by suitable signal aggregation.

These findings make the architecture a credible starting point for the slab sensor, but they do not by themselves establish performance inside a prosthetic liner. Several translation questions remain. First, the force range and boundary conditions of the published test must be compared with the pressures, contact areas, and shear forces expected in the planned slab test. Second, capacitance at the femtofarad-per-newton scale is vulnerable to cable, connector, and environmental parasitics.

Third, encapsulation changes the effective stiffness and therefore the mechanical transfer of force to the dielectric. Finally, repeated flexing, sweat, and temperature variation can change the baseline or produce drift. The value of Albrecht's study is thus twofold: it supplies a defensible transducer geometry and it identifies the calibration problem that the present project must solve after embedding.

2.3 Printed capacitive humidity sensing: Aït-Mammar *et al.*

Aït-Mammar *et al.* investigated an inkjet-fabricated capacitive humidity sensor formed on a flexible polyimide substrate [4]. Their approach used interdigitated electrodes covered by a hygroscopic dielectric, cellulose acetate butyrate (CAB). Interdigitated electrodes are attractive for a liner because both terminals can be patterned on one surface; a separate parallel counter-electrode is unnecessary. The electric field extends through the CAB layer and the surrounding environment. As the polymer absorbs water, its effective permittivity changes, producing a measurable change in capacitance.

The cited device employed 52 electrode pairs with approximately 100 µm gaps and a CAB sensing layer approximately 8 µm thick. The conductive layer was on the order of 300 nm, while the Kapton substrate was approximately 125 µm thick. The study characterised the sensor over a broad relative-humidity range, approximately 20–90%, and reported a clear humidity-dependent capacitance response [4]. This construction is relevant to the proposed slab because it combines a thin substrate, printed conductors, and a solution-processable sensing film. In principle, four small elements can be distributed across the slab to measure spatial rather than single-point moisture conditions.

The literature also exposes a fundamental integration conflict. A humidity sensor must exchange water vapour with the measured environment. Complete encapsulation in impermeable silicone would protect the conductors but would also isolate the CAB layer and suppress or delay its response. The sensing surface therefore needs a vapour-access path, selective membrane, or local exposure to the liner–skin environment. That access route must prevent liquid sweat from shorting the electrodes or causing uncontrolled swelling. Response and recovery times measured in a controlled humidity chamber cannot be assumed to apply under occlusion, contact pressure, condensation, or repeated wetting. Likewise, the published operating frequency and capacitance range must inform the readout circuit; they cannot simply be transferred to an arbitrary capacitance-to-digital converter.

For the present work, Aït-Mammar's study justifies the material principle and planar geometry, not a finished prosthetic sensor. The required follow-on work is to verify CAB deposition on the available printed electrode, characterise film thickness and uniformity, determine whether the proposed opening or membrane preserves vapour access, and calibrate the completed element at skin-relevant temperatures. Hysteresis, saturation, recovery after exposure to high humidity, and cross-sensitivity to bending should be reported. These tests are necessary before the humidity elements can be interpreted as measurements of local liner microclimate.

2.4 Printed resistive temperature sensing: Liew *et al.*

Liew and Lee studied inkjet-printed resistive temperature detectors fabricated from silver nanoparticle inks on flexible polyethylene terephthalate (PET) [5]. The transducer uses the temperature coefficient of electrical resistance: as temperature rises, the resistance of the printed conductor changes. A serpentine geometry increases conductor length within a small area, raising the nominal resistance and measurable temperature-dependent change without requiring a large sensor footprint. The reported pattern was approximately 6.9 mm by 13.2 mm, with a line width near 0.5 mm, spacing near 0.4 mm, and 1.5 mm contact pads.

The study compared two silver nanoparticle inks over approximately 30 °C to 100 °C. A linear relationship between resistance and temperature was reported for both printed conductors. The reported sensitivity was approximately $0.1086\,\Omega\,°\mathrm{C}^{-1}$ for one ink and $0.0543\,\Omega\,°\mathrm{C}^{-1}$ for the JS-B25HV ink [5]. The latter is particularly relevant where the same commercial ink and a Fujifilm Dimatix printer are available. The study therefore provides a direct process precedent for printing four temperature elements on a thin polymer substrate.

Nevertheless, a linear result over 30 °C to 100 °C is not a complete calibration for prosthetic use. The most important physiological changes may occur over a much narrower interval near skin temperature. The readout must resolve small resistance changes within that interval while separating them from lead resistance, contact resistance, mechanical strain, and self-heating. Silver traces on PET are not mechanically neutral: bending and tensile strain can change resistance or initiate cracking, making a nominal temperature signal partly strain-dependent. Encapsulation can improve environmental durability but adds thermal resistance and time lag.

Liew and Lee consequently support the feasibility of the printed RTD, while the present slab must establish application-specific metrology. Each element should be calibrated after printing and again after embedding, against a traceable reference over a skin-relevant range. Thermal response time, repeatability, drift, bending sensitivity, and agreement between the four elements should be reported. A four-wire measurement would reduce lead-resistance error where space and interconnect count permit; otherwise, conductor and lead resistance must be included in the calibration model.

2.5 Discrete temperature measurement and thermal control: Ghoseiri *et al.*

Ghoseiri *et al.* developed a temperature measurement and control system for transtibial prostheses using 12 temperature sensors distributed on opposing sides of the interface [6]. Their work connects temperature measurement to a clinically meaningful mechanism: heat accumulation promotes perspiration, and the combined warm and humid environment can affect comfort, skin condition, and prosthetic suspension. The study also shows the value of spatial measurements. A single temperature value cannot represent gradients created by local loading, anatomy, material thickness, and nonuniform heat transfer.

The system did more than log temperature. Measured differences were used to control heating or cooling, demonstrating a closed-loop thermal-management concept. Its complexity, however, also reveals the practical limits of adding components to a wearable socket. Sensors, conductive paths, control electronics, and thermal-management hardware add stiffness, volume, power demand, and possible failure points. Ghoseiri *et al.* identified further miniaturisation and system refinement as prerequisites for practical translation [6].

The present slab uses two TMP36 integrated-circuit sensors as discrete internal references. According to the manufacturer, the TMP36 operates from 2.7 V to 5.5 V, has a nominal output slope of $10\,\mathrm{mV\,°C^{-1}}$, and is specified over -40 °C to 125 °C [8]. Its simple voltage output and low current make it convenient for prototype acquisition. Two sensors can indicate whether a thermal gradient exists across the slab and can provide an independent comparison with the printed RTDs.

The TMP36 choice is experimental rather than automatically liner-compatible. The TO-92 package is substantially thicker and stiffer than a printed trace. Within a 3.5 mm middle layer, a package approximately 2.8 mm thick leaves little material above and below it. Local stiffness and stress concentration must therefore be assessed. The sensors measure their package temperature, not instantaneous skin temperature, and the surrounding silicone introduces lag. Ghoseiri's work supports distributed temperature measurement, while the TMP36 datasheet supports the electrical choice; neither source proves that this package will remain comfortable or mechanically safe inside a final liner.

2.6 An instrumented liner precedent: Paternò *et al.*

Paternò *et al.* provide the closest system-level precedent for the present project [7]. They developed a custom transtibial liner with 12 embedded Hygrochron temperature-and-humidity loggers and compared its microclimate measurements with those obtained using a participant's conventional liner. The liner was designed from the user's geometry, and sensor locations were incorporated through a controlled manufacturing route that included machining. The work demonstrates that distributed temperature and relative-humidity measurement can be integrated into a wearable liner and used during a structured protocol rather than only in a benchtop chamber.

The study is important for three reasons. First, it frames temperature and humidity as spatially varying quantities. Paternò *et al.* reported mean conditions at the start of one protocol of approximately 31.42 °C and 68.13% relative humidity, with values changing during rest and activity [7]. Second, it demonstrates a route from user geometry to instrumented liner manufacture. Third, it acknowledges that the sensor must be integrated without undermining the liner's primary mechanical and comfort functions.

Its limitations are equally instructive. The Hygrochron devices are rigid packages and are much larger than printed thin-film elements. The work was an early case study involving a limited participant sample and a controlled protocol. It does not establish long-term durability, washability, or universal comfort. Nor does it measure normal pressure or shear, so the recorded microclimate cannot be directly related to local mechanical loading. These limitations define an opportunity for the present design: printed elements may reduce thickness and increase spatial resolution, while a multiaxial capacitive element can add the mechanical variables absent from the instrumented-liner precedent.

The proposed slab should not be presented as a direct miniaturisation of Games's liner. It combines sensing mechanisms with different substrates, readout requirements, environmental sensitivities, and mechanical stiffnesses. The slab is instead an integration experiment intended to determine whether those mechanisms can coexist within a silicone laminate. A flat slab is an appropriate preliminary specimen because layer placement, encapsulation, electrical continuity, cross-sensitivity, and local compliance can be examined before undertaking the more expensive and less repeatable manufacture of a full curved liner.

2.7 Synthesis and research gap

No single reviewed study supplies the complete solution required here. Albrecht *et al.* provide a thin capacitive architecture capable of separating normal and tangential loading, but not a validated prosthetic-liner implementation. Aït-Mammar *et al.* provide a printable humidity-sensing mechanism, but integration must preserve vapour access. Liew and Lee provide a printed silver RTD process, but it requires calibration near skin temperature and separation of thermal and strain effects. Ghoseiri *et al.* show why distributed thermal measurement matters, but also illustrate the bulk and complexity of active socket systems. Games *et al.* demonstrate that temperature and humidity can be logged from an instrumented liner during use, but their rigid packages do not provide pressure or shear data and limit miniaturisation.

The resulting research gap is not merely the absence of another sensor. It is the absence of a compact, mechanically integrated test article that combines:

(i) directional mechanical sensing;

(ii) distributed printed temperature measurement;

(iii) distributed printed humidity measurement;

(iv) discrete temperature references; and

(v) a silicone layer structure representative of a liner.

The proposed 56 mm by 55 mm by 5.2 mm slab addresses this gap at prototype scale. Its three layers comprise a 0.20 mm first layer, a 3.50 mm sensor-containing middle layer, and a 1.50 mm cover layer. The planned sensing set comprises one four-capacitor normal/shear element, two TMP36 devices, four printed RTDs, and four printed humidity elements.

This literature supports the architecture as a research hypothesis, not as a validated result. The decisive contribution must come from fabrication and testing. The completed slab must be evaluated for dimensional fidelity, electrical continuity, local stiffness, normal/shear decoupling, temperature accuracy, humidity response, response time, repeatability, drift, hysteresis, and behaviour after bending or cyclic loading. Only then can the design be judged suitable for progression from a flat integration coupon to a curved prosthetic liner.

3 Methodology

4 Results

5 Discussion and Conclusion

References

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